ADITYA COLLEGE OF ENGINEERING AND TECHNOLOGY

No 2/9, Kamakshipura, Hesaraghatta Hobli, Yelahanka, Bengaluru- 560089
Affiliated to VTU, Belagavi | Approved by UGC/AICTE/Govt. of Karnataka, Bengaluru

DEPARTMENT OF PHYSICS

ASSIGNMENT – II

Academic Year	2024-25	**Dept.**	Physics
Program	BE	**Date of issue**	23-05-2025
Year / Semester /Sec	2025/II/A/B/C	**Date of Submission**	31-05-2025
Course Code	**BPHYS202**		
Course Name	Applied Physics for CSE Stream		

Note	***1. Answer all the questions***

Q. No.	Questions	Marks	COs	RBT level
1	Explain Expectation value, Eigen function and Eigen values.	08	CO 2	L3
2	Assuming the time independent Schrodinger wave equation, discuss the solution for a particle in One-dimensional potential well of infinite height. Hence obtain the normalized wave function.	10	CO 2	L3
3	Discuss the Wave functions, probability densities and energy eigen values for a particle in a box with diagram. [Assume normalized wave function of the particle.]	10	CO 2	L2
4	Explain Moore's law and its end.	06	CO3	L2
5	Explain the representation of qubit using Bloch sphere.	06	CO3	L2
6	Elucidate the difference between classical computing and quantum computing.	06	CO3	L2
7	Explain the matrix representation of 0 and 1 states and apply identity operator to $\|0\rangle$ and $\|1\rangle$.	05	CO3	L2
8	State Pauli Matrices. Apply Pauli matrices on the states $\|0\rangle$ and $\|1\rangle$.	08	CO3	L2
9	Explain Hadamard Gate. Apply Hadamard Gate on the states $\|0\rangle$ and $\|1\rangle$.	07	CO3	L3
10	Explain orthogonality and orthonormality with an example for each.	05	CO3	L2
11	Given $A = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$, prove that $A^{\dagger} = A$.	05	CO3	L3
12	Explain the working of phase gate mentioning its matrix representation and its truth table.	08	CO3	L3

Prepared by:

Prof. Nayan Kumar H N
Assistant Professor
Dept. of Physics

Prof. Pavendhan M
Associate Professor
HOD